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SEPIA: A REAL-TIME EXPERT SYSTEM THAT AUTOMATES TRAIN ROUTE MANAGEMENT[1]

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Abstract. The SEPIA system is a real-time software package that handles all non-safety-related activities usually carried out by a signalman, mainly the selection and setting of train routes, the management of route conflicts, the choosing of the station platform and parking tracks, and the follow-up and checking of train movements. As an added advantage it processes and distributes train traffic information for management needs and informing customers. Its major aims are to boost the productivity of signal men, to upgrade the quality of service with more emphasis on consumers, and to optimise overall traffic over large areas. Its decision component stems from AI techniques.

Key Words. Rail traffic; real-time computer systems; artificial intelligence; process control; expert systems; optimisation; heuristic programming; human factors

1. INTRODUCTION

As in all guided transport systems, a train that runs on the railway network system can only follow its assigned track. A train driver cannot, by himself, overtake a train that limits his train speed, select one road from the set of available paths of the railway network, nor define the rules of priority at junctions. He can only choose the train speed, which is subject to the signal guidance information, and the braking efficiency of his train. All the route selection and command activities are performed at ground level by men dedicated to managing the train traffic. This is a hierarchical two-stage activity shared between the signalman and the train dispatcher, who acts as a traffic supervisor. For technical and historical reasons, the railway network is divided among a number of signal boxes, responsible for the choice and order-setting of routes. In the early days of rail transportation, this work was done by the local control of each switch on a route. Later, the local control was transferred, and centralised in signal boxes. This centralisation reduces the signalman's duties, and it allows one man to work on many switches and routes from a single location. Now, in modern signal boxes, switching devices are remotely controlled and managed by computers. This activity is accomplished under the supervision of a train dispatcher who manages several signal boxes. He is responsible for strategic decisions on traffic management, such as the train priority range on the main lines. Thus, he determines the signal-box exploitation constraints for the signalman who, under these directives, optimises the management of traffic in the area for which he is responsible.

1.1. Domain Problem

As in most cases of industrial activities dealing with process control, the management of railway traffic is done in two steps. One or several days before a train runs, a planning document is produced by certain specialised services, and its content is used as a reference framework for performing the traffic control. At running time, prior to setting a command, an operator modifies these planned directives, if necessary, to take account of unexpected data changes such as unavailability of a route, unscheduled running, the break-down of a train or of some railway network devices such as

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switches, signals or relays. The SEPIA system executes, automatically and under its own control, these functions of checking the validity of the planned directives, adjusting them if necessary, setting the commands and controlling the results.

1.2. Historical Background

In 1991, a prototype version of this system was linked to a small signal box and successfully tested. A full-scale version of SEPIA, with industrial properties, was subsequently developed and linked up at the end of 1994 to the signal box of the Le Mans station, a strategic railway junction node in charge of 400 routes that serve six main directions. The traffic within Le Mans is a heterogeneous system of trains: high-speed (TGV), main-line, regional, freight, and service trains. This diversity provides a rich panorama of situations for the SEPIA experiment. The Le Mans experiment will be completed by mid-1996. During this period the SEPIA system will be continuously enriched by the introduction of new expert rules and the consideration of demands concerning user needs for information retrieval and presentation. When this study is concluded, the SEPIA automation function will be integrated into the new generation of signal-boxes.

Later on, SEPIA will become part of a new system for dispatchers who supervise intensively used railway lines, especially in Parisian suburbs. This new product will be responsible for the overall traffic management. Related constraints on resources, including driver, commercial controller, locomotive and train car reuse planning will take account of possible repair solutions in the event of something going wrong. Indeed in many cases, a reasonably effective proposal is to suppress a train's circulation or limit a train's stopping pattern. Such a solution may entail a future strain on certain resources if their reuse is disrupted. For example, the driver of the cancelled train will be unable to reach the place where the next train that he is scheduled to drive is stationed.

1.3. A Process-Control Problem

The canonical frame of an industrial process control is constituted of an endless loop which achieves the following operations: to get information from the controlled process, to compute and to optimise a target function with these data, to select and to set commands to reach the computed goal, to analyse the output information and to verify how the commands were achieved.

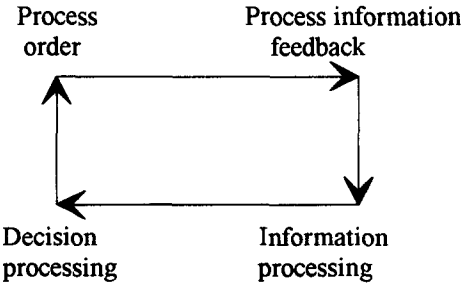


Fig. 1. The endless loop of process control

For complex process control, the tasks of computing, optimising, selecting and ordering are essentially within the operator's sphere of responsibility. In SEPIA, these tasks are automated thanks to artificial intelligence techniques.

1.4. SEPIA Presentation

The architecture of the SEPIA system is based upon the robot paradigm. Therefore, the overall problem has been similarly divided into five subsets: 1) information capture, 2) mapping captured data into a railway network representation, 3) planning, 4) task execution, 5) order control.

In this paper, the third point will be presented in more detail because the planning task is the core of the SEPIA system. The expert systems that direct the planning search are included in this task. The planning process can be formulated as an allocation problem of reusable resources. The definition of the concepts adopted to describe this problem is explained first. The role and interest of the SEPIA expert systems in efficiently performing this activity will subsequently be presented.

2. FORMAL PLANNING MODEL

2.1. Problem Definitions

Resource. With minor simplifications, a route r_j is an oriented arc of the railway network. It comprises a set of ordered smaller parts (the track-circuit zones) z_k, such as: $r_j := \Sigma_k z_k$.

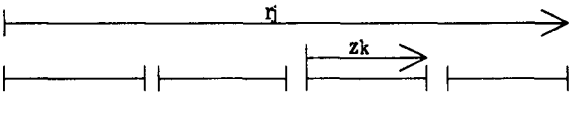


Fig. 2. Route r_j with its parts z_k.

Different r_j may or may not share the same z_k, i.e. in general $r_{j1} \cap r_{j2} \neq \varnothing$. The allocation of an r_j to a task t_i is exclusive, but t_i makes gradual use of r_j

and frees the z_k in a sequentially ordered manner. When r_j is allocated to a t_i the use of some other $r_{j''}$ is forbidden, in order to prevent the trains from cutting-across routes or from colliding at the points. The lock-up period of time of one r_j and of its related z_k and also by transition $r_{j''}$ depends on the t_i task class. A resource may be used in two modes, first to move t_i from one point to another, second as a stationary t_i place. According to the selected mode, additional properties of r_j are needed, and may, to a certain extent, restrict the choice. For example, for the first mode the transit duration, or for the second, the platform length or the passengers' accessibility and convenience are properties.

Task. A train that travels through the area under the control of a signal box can be seen as a task t_i that uses some resources r_j of the railway network. This task is responsible for accomplishing the stopping pattern of a train that defines its production constraints. A stopping pattern is a list of time-stamped train movements from one point to another. For example, t_i enters the signal-box area at d_0, then stays in the railway station S_1 from d^a_1 until d^d_1 and leaves the signal box area at d_2. It is generally possible to have many booked stops of a train at different stations.

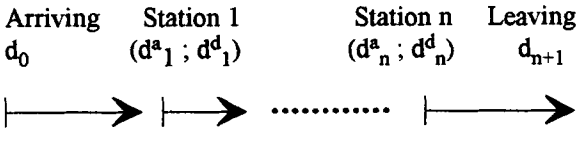


Fig. 3. Train stopping pattern

The train trip may or may not be strongly constrained at special locations such as arrival and departure points and intermediate stops. Frequently, there are many different potential paths joining two successive points. Each of the potential paths p_t is a series of successive routes such as: $p_t := \Sigma_i\ r_i$. In other respects, the choice of a particular value in the set of potentials for the special points may be made freely, but in general not at the same cost depending on the selected value.

The freezing of resources, for day-to-day track maintenance or for failures, can also be seen as tasks.

Target Function. This function measures the cost of an allocation of resources for each train that runs and also for all the trains expected to run on a particular day. This last and overall measure takes into account that the choice of particular resources for one t_i may lead to an irrelevant choice for another $t_{i''}$ because there are only bad resources left free for $t_{i''}$. The cost of a solution can be perceived according to several criteria. In SEPIA two main criteria are taken into account. The first roughly measures the customer discomfort associated with an allocation of resources. The second is a similar measurement expressing the trouble caused to SNCF (the French national railway company). The customer indicator shows, for example, the inconvenience caused by allocating an irrelevant platform for one train. This resource may be ill-suited for many reasons, such as: too low a platform would make the ascent and descent of passengers difficult; an unusual platform would make the distance between the train and the station exit longer, or when a different platform is assigned for the connecting train. The SNCF criterion measures train delays resulting from the temporary unavailability of a resource. Two types of drawbacks may result from this occasional strain on resources. First, a train delay may ruin the overall planning for the reuse of rolling stock and drivers. This will increase the production cost and jeopardise the regularity of traffic over time. Second, the service quality will be damaged, and in the end lead to customer dissatisfaction and the eventual loss of clientele. Like the customer criterion, it is necessary to take into account the delay for one train and the cumulative delay generated.

All these measurements are empirically based, and a great deal of work remains to be done in order to specify a good tool, with a proper theoretical basis, that will facilitate the evaluation of the quality of traffic management in both automatic and manual modes.

Graph of Problem Resolution. There are three distinct ways of representing and considering problem-solving tasks (Simon, 1983). In SEPIA, one of them, constraint satisfaction, is not used. The last two are briefly presented.

1. A state graph. The space of the sequential states of a problem resolution is constructed as a graph in which nodes represent states and each node-link represents the transition from one state to the next. In general, this graph is a tree. A state is a data structure that contains information about the current step of the resolution. In the SEPIA planning task, a state is an attribution of a set of resources r_i to tasks t_j. If n tasks must be planned, the state graph looks like Fig. 4.

   The triangles represent local states for the task t_{j*}, when previous tasks t_j, such as $j < j^*$, are fixed. At each triangle end-node, a state St_{i*} is an allocation of resources for all the t_i such as $i \leq i^*$.

$$St_{i*} := \{g(t_1) ; g(t_2) ; \ldots ; g(t_{i*})\}$$
$$\text{where} \quad g(t_i) := \{r^{ti}_{i1} ; r^{ti}_{i2} ; \ldots ; r^{ti}_{ii}\}$$

$g(t_i)$ is the list of resources assigned to t_i that will allow t_i to ensure its stopping pattern. An r^{ti}_i attribution is time-stamped and in fact ties up more resources than the single r^{ti}_i, as has been indicated in the resource definition. An entire solution St_n synthesises the attribution of resources for the n trains studied, and for the selected path graph.

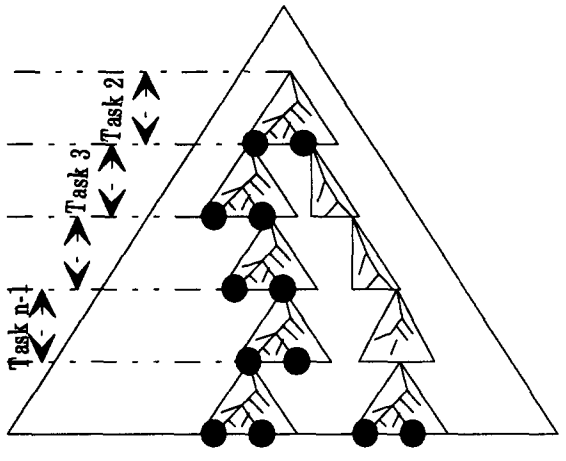


Fig. 4. Diagram of the state graph.

2. An OR-AND graph. This graph illustrates the decomposition problem as sub-problems, and the interrelationships between the sub-problems. In SEPIA, this formalism is only adopted to treat situations where a train traffic pattern is very constrained on an intermediate station. In such a case the train's resource allocation can be divided into two similar sub-problems, one to find resources from the intermediate point to the arrival point and the other from the intermediate point to the departure point.

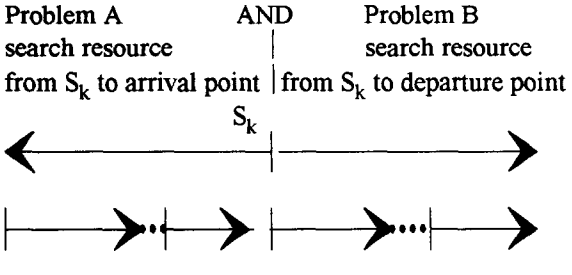


Fig. 5. The AND decomposition problem

2.2. Graph Exploration

All scheduling problems (Fox, 1990) result in a state graph exploration, but when the search space is very large it is necessary to find an efficient search algorithm for controlling the order of application of the operators.

Basic Rules Principles. First, an assumption is made according to the principle that the most constrained task is treated first. That consists in selecting the task that has the fewest choices to fix the resource values. Second, trains are scheduled some time before their running through the signal box area. The anticipated scheduling is made for an unspecified period of time. For example, the time gap value depends on the observed context of train traffic. These anticipated periods are calculated in such a manner as to avoid loss of time from computing with unreliable time data about train arrival. Third, resource allocation is made for a set of trains that have some interrelated constraints that condition the choice of resource for each of them. Fourth, the backtracking is limited, as far as possible, to trains currently under study. These approaches are implemented via expert systems and specialised functions that define the search strategies.

Operators and Pruning techniques. The search for solutions is made in two stages. Following Fig. 4, the first step corresponds to the vertical classification of tasks, and the second to the exploration of interior triangles. For level one, the search is guided by an expert system that arranges trains in the most constrained order. It also analyses search dead-ends and directs backtracking either by selecting the locking train, or by reformulating the submitted problem, or by accepting a train delay. Inside a triangle, the search is conducted by strategy operators.

The three most frequent are briefly presented.

The depth-first operator proceeds sequentially from the arrival to the departure (or terminal) train location. It moves from one sub-goal to the next. It first tests the value indicated in the planning document. If the cost of use is lower than the authorised cost threshold, this assignment is accepted. If not, either a breadth search is conducted in the sub-goal until a cost-acceptable resource is found, or a backtracking to the previous sub-goal is directed.

The breadth-first operator selects the best resource for each sub-goal, still conditional on the value chosen for the previous sub-goal.

The directed operator is driven by a dedicated expert system that defines suitable and/or substitutable resources for each sub-goal.

3. USE OF ADVANCED TECHNIQUES

3.1. Expert Systems

The operator model that mimics the operator's decision-making process contains all the heuristics and rules of thumb used by experts in traffic management. This knowledge is written as production rules. There are currently about three hundred rules into the SEPIA knowledge base. This number will tend to increase as the experiment goes on. The knowledge base is structured in specialised expert systems. For example, one of these systems is in charge of commercial rules and service quality. It deals with flexible constraints such as: how and when to impose or to relax constraints for selecting the departure or the train arrival platform; to optimise the choice of the platform for connecting trains from the passenger's viewpoint, or to manage and minimise, if possible, the drawbacks due to opposing flows of passengers on the same platform. Another expert system is responsible for managing unsolved conflicts between trains, and allows priority to be given to one particular train.

The state graph exploration techniques, previously described, are also incorporated into specific groups of rules. More details are given below about the implementation of the basic rule principles.

First, the starting date of the planning process for a particular train is set. This date is determined according to the properties of trains expected to run into the area under control. For example, a train of unusual length needs to be scheduled a long time before its arrival because it may only use a platform with the correct length. It is also better to reserve that single resource a long time in advance to avoid it being attributed to a train without such a length constraint. Similarly, a train without any distinctive constraint can be scheduled a short time before its arrival because there are a lot of replacement resources that may be assigned to it. This rule-based approach to fix the planning anticipation value can be seen as a flexible and dynamic combination of a fixed planning anticipation time and a last moment planning when a train arrives in the signal-box area.

Second, when a particular train is planned it is recommended that all the trains connected with it be scheduled in the same run. There are many types of links between trains. Some are material ties; typically, a train that finishes its mission in the area will later run under another name. When a location is chosen for the first train terminus, this choice will implicitly be the departure location for the second train. Other links are not so clearly established and are described by rules. For example, trains that run within close periods of time may be in competition for routes and platform resources. To deal with this situation, they are placed in the same group of trains to be scheduled together. In this way, all the closely linked trains are scheduled simultaneously to take into account their mutual constraints in terms of the process of resource allocation.

Third, the group of trains is arranged according to a more constraint-first classification criteria. The appreciation of the constraint level value for each of the trains in the group is an expert-system task described by rules such as the following: a service train has less priority than a TGV with passengers and may have a lower-quality solution. It may, therefore, be planned at a lower rank of the group planning process. This unfortunate position on the waiting list for resources distribution may lead it to use the left-over resources that may be ill-suited for the running of the train.

Fourth, the objective of process management is not imperatively to obtain the best solution, but rather to get a good solution, the best if possible, within a specified duration of computing. To this end, an expert system is dedicated to specifying the minimal objectives to be achieved by the planning task, according to the operational context. The evaluation cost of a particular solution is very subjective and depends heavily on the context of traffic management. When the traffic management is sound without any delayed trains, it is normal to seek a very good solution. However, when there are numerous delayed trains or when there is a breakdown of any railway network device that limits the number of usable resources, a good solution may be very costly. To deal with this problem, the maximal cost level to be reached for an acceptable solution is deduced from rules based on the traffic management context. These rules define the threshold value of a "not too bad" solution that may be retained as the solution. If the search-path algorithm cannot find a solution satisfying this value, reference is made to specialised rules that manage unsolved conflicts by defining the train priorities. The SEPIA system also uses an economic function to calculate a solution cost. It is a multi-criteria function with variable coefficients to balance each of them. Up to this point, only two criteria are considered. The first is to measure the SNCF cost of a solution as a sum of the train delays. The second is to measure the consumer inconvenience caused by this solution. For example, when the train's stopping platform differs from the usual one, there is a cost to the regular consumer who may omit to consult the information panel. Specific rules indicate the cost of each type of

potential consumer discomfort in this solution. However, there are other rules that specify the weighting of the SNCF and consumer criteria. The formula for calculating the cost of a solution is:

$$\text{Solution cost} = \Sigma \text{SNCFcost}_j * \alpha_{SNCF} + \Sigma \text{Consumercost}_k * \alpha_{consumer}$$

where: α_{SNCF} = SNCF criteria weight
$\alpha_{consumer}$ = Consumer criteria weight

These two coefficients facilitate a fine-tuning of the acceptability of a solution. When the traffic is severely disrupted $\alpha_{consumer}$ and α_{SNCF} tend to zero. When the traffic is normal $\alpha_{consumer}$ and α_{SNCF} tend to one; in a combined situation with late trains, $\alpha_{consumer}$ and α_{SNCF} may vary between these two boundary values.

3.2. Knowledge Representation

The overall aim of a train, which consists of completing a commercial mission according to a specified time schedule, can be reduced by applying knowledge to a set of sub-goals, each of them being more tractable than the overall one. In SEPIA, trains that have similar mission patterns fall into the same train class. One class defines the decomposition framework in sub-goals and indicates the related default arguments. This class information must be used first. The path linking two successive sub-goals is calculated by a branch-and-bound type of algorithm. This first exploits all the available information such as the default path between two railway network places. To facilitate the search, the different paths joining two places are preferentially ordered, especially from a transit-time viewpoint.

The description of a new train stopping pattern is made in a declarative way. This constitutes a powerful tool to guide the resource allocation engine. It is very easy to modify or to add new patterns and the SEPIA model may, therefore, be extended to unexpected structures of train travel in the signal-box area. This is also a means of introducing SEPIA's parameters. It will be used when new signal-boxes are controlled by SEPIA.

3.3. Robotics

As a result, the planning task formulates action assignments into an action plan. In SEPIA, an action is an order that will be set. Each order is conditioned by a list of conditions to be checked before being sent. For example, a condition may be a timer, or a particular occupation position that becomes true when a train arrives at a specific location, or for a second train when the first train's movement depends on the second train's position. A task is attributed to plan execution, and continuously monitors condition satisfaction. When all the conditions of an action are true, the order is set and an acknowledgement task is issued to ensure the completion of the order within a given time.

3.4. Reactive System

In the real world, unexpected environmental changes may invalidate the preformulated plan. When such a failure occurs there are two ways to handle the situation: replanning or recovery (Chang, *et al.*, 1993). In the SEPIA system a combined approach is adopted and a partial replanning is done to repair the former plan. Dedicated rules give directives to the planning task, such as circumscribing the set of trains to be planned again, or increasing the threshold cost of an acceptable solution.

4. SOFTWARE PRODUCTION PROCESS

The software production process was a research challenge in itself. The main goals were to obtain a sturdy and evolutionary system, composed of reusable by-products and software components.

4.1. An Object-Oriented Model

To reach these targets, the SEPIA software is strongly object-oriented and written in C++ language. The expert systems are written with the ART-IM™ package. Every step of the software development is conducted with OBJECTEERING™, a CASE tool supporting the CLASS-RELATION™ method (Desfray, 1994). That methodology is very similar to the well known OMT method. Many class-libraries were produced and reused within different SEPIA modules, and even in other projects.

4.2. Modelling and Software Design

SEPIA is a model-based system. It is composed of four main models: a model based on a deep knowledge representation of the railway network which is under its control, and three other models concerning the human operator, the train and the planning function respectively. The time graph of the resource availability is attached to the railway network representation. Each of these models is

implemented as co-operative, independent and dedicated UNIX tasks. They are linked by a software bus and they communicate by message transmission. In addition, this software architecture provides the opportunity, when necessary, to possibly reduce computing time by transferring the SEPIA system to a distributed computing architecture. A diagram of the SEPIA architecture is put at the end of the paper (Fig. 6).

4.3. Validation of SEPIA's Functional Availability

Prior to connection to real-world traffic management, the SEPIA system has been tested intensively with SISYFE, a train traffic simulator (Chardonnal *et al.*, 1994). Thanks to this system, the functioning quality, especially when applied to difficult cases, can be guaranteed and demonstrated to the user with no hindrance to traffic punctuality. This approach will improve the end-user confidence in the SEPIA activity.

5. HUMAN FACTORS

The high level of automation generated by SEPIA will drastically change the way signalmen work. This system will lead to an increase in the sizes of the domains that they supervise. The extension of areas will itself generate more possibilities for optimising management decision making.

In the future, the signalman will act more like a dispatcher, than like a traditional signalman who takes care of a switching device. Nevertheless, he will always be in charge of handling the safety protocols. To bridge this important cultural work transition, a study related to the SEPIA project was conducted, to discover the new aspects of human factors in traffic management.

5.1. Four Automation Levels

To take into account the conclusive recommendations of the above-mentioned study, SEPIA was designed to work according to four, different and gradual, automation levels.

Vigilant Mode. In this functioning mode, SEPIA listens to the man-made traffic management. The SEPIA railway network software image is updated when information is received from the operator's keyboard or from receptors that locate trains on the railway network. At any time, SEPIA must be able to assume either the decision-making or the automatic mode.

Decision-Making Mode. SEPIA assumes control of the traffic management, but important decisions must be confirmed by the man who supervises the SEPIA system.

Automation Mode. SEPIA is fully in charge of the traffic management. In important signal boxes there are many station operators, each of them supervising the traffic of a specific part of the whole signal box area. SEPIA can automate some of these stations and stay in vigilant mode for the other ones. This characteristic property enables the operator to continue using his/her professional knowledge.

Incident mode. This mode is used when an incident, like a device failure, obliges safety orders to be set. In this case, the SEPIA supervisor sets the borders of the area concerned in the safety problem. In this zone, he will assume responsibility for decisions taken and set orders, but he may delegate the traffic management over the complementary unaffected area to SEPIA.

5.2. Human-Computer Interfaces

In addition, a major effort was made to create innovative user interfaces. The main targets of this work were to explain the choices made by SEPIA, and to illustrate the forecasting of SEPIA choices.

Both results are formulated using graphs in order to communicate information. Three main graphics are proposed: the train diagram, the track-occupation diagram and the railway network diagram. For each graph, it is possible to visualise, with characteristic drawings: 1) the theoretical train movements, as indicated in the planning document, 2) the use of resources planned by SEPIA, 3) the circulation of trains observed.

5.3. Supervisor

Paradoxically, in almost fully automated process control as targeted with the SEPIA system, the supervisor's role is essential and it is strongly recommended (Bisseret, 1985) to place tools at the supervisor's disposal. These tools should enable him to analyse past problematic situations, or to consider in advance how to forestall similar situations before they happen. To this end, the SISYFE simulation system will be delivered with the SEPIA package. With SISYFE, the supervisor will be able to maintain and consolidate his professional knowledge, and will take self-training courses for unusual problems. It will thus be possible to specify new rules in order to progressively improve SEPIA.

6. CONCLUSION

The SEPIA system will boost the productivity of signalmen, improve working conditions in the handling of technical hitches in train-traffic and increase the quality of traffic management, particularly when a train, a switch or a signal breaks down. In addition, it will ease the transition from a situation where safety was essential, to a situation with the same safety level but with more emphasis on the traffic management optimisation and on the improvement of the service quality. This system permits the traffic management to be extended over very large areas. This makes traffic optimisation possible, by giving a wider choice of alternative tracks. At the present time, SEPIA can handle a normal day's train circulation (250 trains) but it still needs to be improved to work well when a major breakdown occurs. The latest nine-month experiment will be used to reach that target. From a software development viewpoint, the use of advanced computer techniques like expert systems was justified for two seemingly contradictory reasons. First, AI techniques are recommended to solve difficult and ill-defined problems that cannot be treated by algorithmic methods. Second, one of the aims of this project was to produce a robust (since it continually runs in real-time) but also alterable (for evolutionary purposes) software package. In SEPIA, the rule-based techniques are adopted for these two reasons. These techniques allow the implementation of heuristics and rules of thumb used by traffic management experts as well as the gathering, in this single module, of functions that could evolve continuously, and that should support the functional maintenance.

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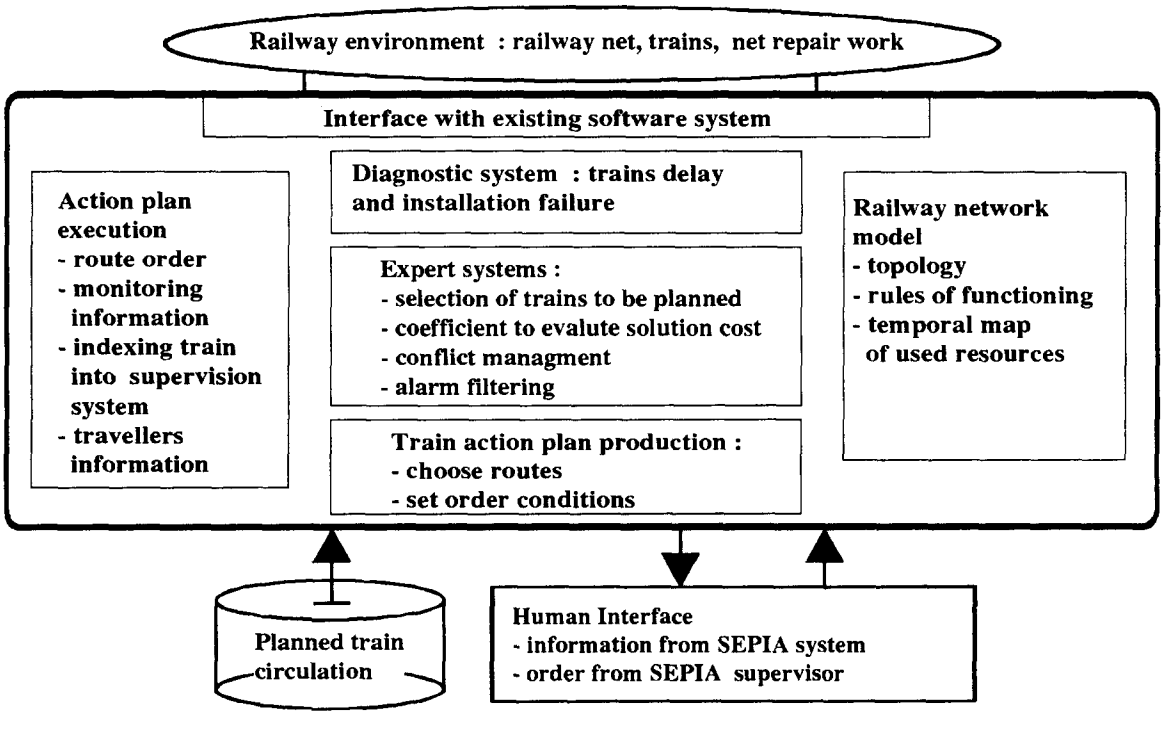


Fig. 6 The SEPIA architecture.